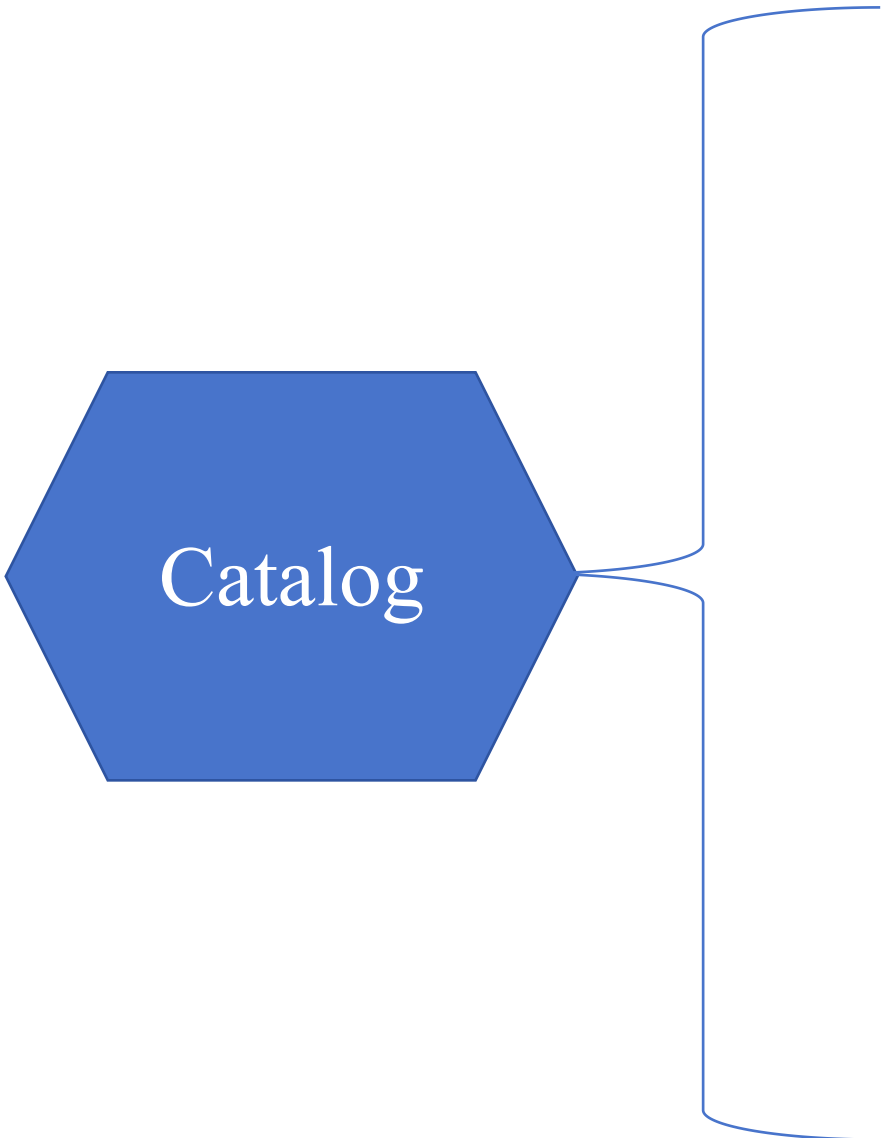


Zintilon Co., Limited



Sheet Metal Processing Capability

IATF16949 | ISO13485 | ISO14001 | ISO9001 Certified | zintilon.com | Shenzhen · China



Catalog

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1. Sheet Metal Equipment List

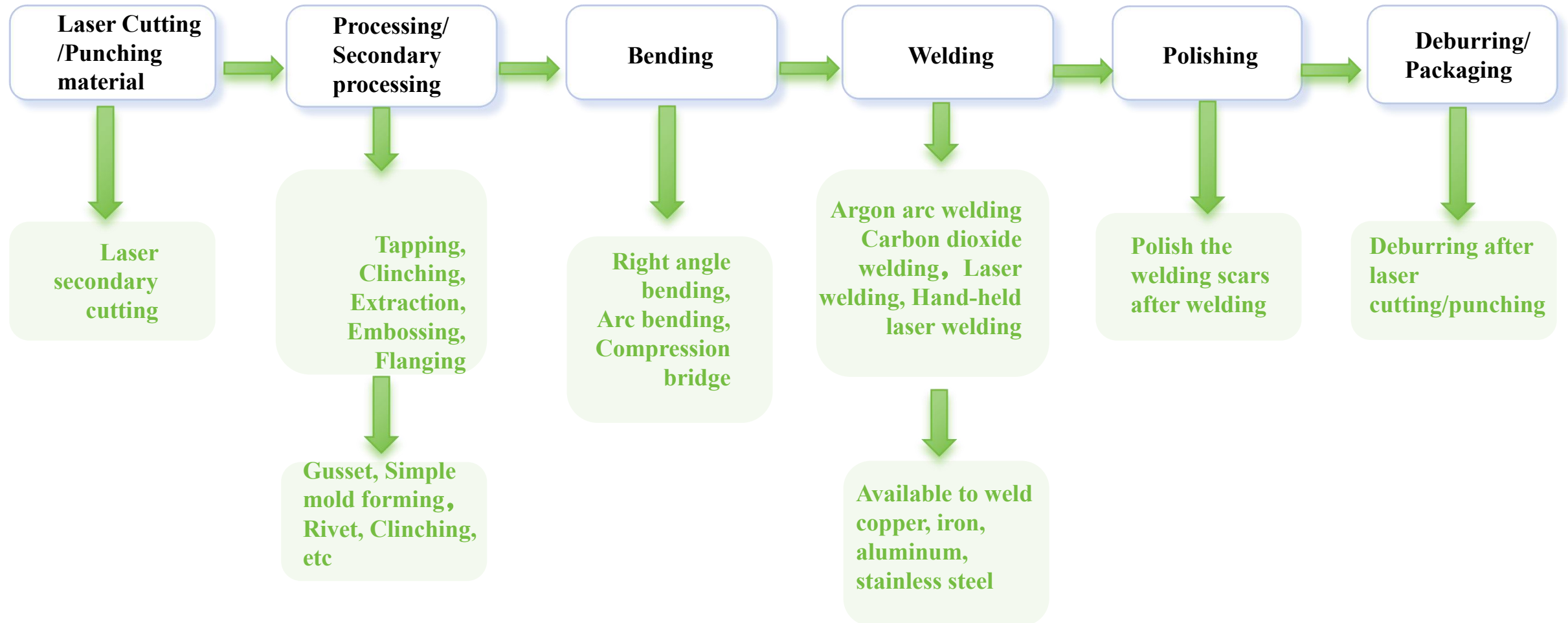
Sheet Metal Equipment List							
No.	Equipment	Brand	Qty	Unit	Processing Range	Machining Capability	Note
1	Laser Machine	Eeike MK3015F (3000W)	1	Set	1500*3000mm	Iron:15mm/ Aluminium alloy:8mm/ Stainless steel:10mm/ Copper:3mm/ Brass:Within 6mm (Aperture accuracy: ±0.05; Trimming accuracy±0.1)	
2	Laser Machine	Dazu MPS3015C(6000W)	1	Set	1500*3000mm	Iron:25mm/ Aluminium alloy:16mm/ Stainless steel:20mm/ Copper:6mm/ Brass:Within 12mm (Aperture accuracy: ±0.05; Trimming accuracy±0.1)	
3	Bending Machine	Yike ES3512/EG6520/ES1003	6	Sets	1.2m-3m	35T-100T, within 3m (Bending accuracy ±0.2)	
4	Bench Grinders	Xihu S3ST-250	1	Set		Drill sharpener	
5	Bench Tapping Machine	Xihu MODEL-SWJ-16	1	Set		Tapped hole: within M16	
6	Bench Drilling Machine	Xihu MODEL-Z516A	2	Sets		Counter sink within M16	
7	Riveting Machine	Haiji H618	2	Sets	6T	Press riveting, hole extraction, bud extraction	
8	Oil Press Machine	HTES-300SSB	1	Set	900mm*1200mm*300T	Convex hull forming within 900*1200mm	
9	Pull Riveting Guns		1	Set		Rivets within M14	
10	Pull Nail Guns		1	Set		Sheet meta parts with nail assembly type	
11	Laser Welding Machine	3000W 21T	1	Set	3000W	Aluminium alloy/ Stainless steel/Iron/Brass: Within 8mm	
12	Carbon Dioxide Gas Shielded Welding Machine	AFT-300	1	Set		High efficiency, can weld large welding products	
13	Argon Arc Welding Machine	BEM-35011	1	Set		Aluminium alloy/ Stainless steel/Iron/Brass: Within 10mm	
14	Manual Wire Brushing Machine		1	Set			
15	Sander		4	Sets			
16	Altimeter	0~700mm	1	Set		Precision Measuring Instrument	
17	SanFeng Digital Altimeter	192-664-10	1	Set		Precision Measuring Instrument	

2. Sheet Metal Raw Material List

Sheet Metal Raw Material List

Aluminium Plate				Stainless Steel				65Mn		Cold Plate SPCC		Copper					
AL5052		AL6061		SUS304		SUS301		Specification	Thickness	2500*1250		T2 Copper					
Specification	Thickness	Specification	Thickness	Specification	Thickness	Specification	Thickness			Specification	Thickness	Specification	Thickness				
2440*1220	0.2	2440*1220	1.0	2440*1220	0.4	2440*1220	0.15	500*270	0.3	2500*1250	0.5	1500*600	0.5				
	0.3		1.2		0.5		0.2	640*500	0.5		0.6		0.8				
	0.5		1.5		0.6		0.25	2440*1220	0.8		0.8		1.0				
	0.6		1.6		0.8		0.3	300*500	1.0		1.0		1.2				
	0.8		2.0		1.0		0.4	300*800	1.2		1.2		1.5				
	0.9		2.3		1.2		0.5	600*500	1.5		1.5		1.6				
	1.0		2.5		1.5		0.6	1000*300	1.6		1.6		2.0				
	1.2		3.0		1.6		0.8	2000*300	2.0		2.0		2.3				
	1.3		3.2		2.0		1.0	500*500	2.3		2.3		2.5				
	1.5		4.0		2.3		1.5	500*500	2.5		2.5		3.0				
	1.6		6.0		2.5			1220*400	3.0		3.0		3.2				
	2.0		7.0		3.0			1000*610	3.2		3.2		3.5				
	2.3		Brushed Plate				3.5	SUS316			Electrolytic Sheet SECC			3.5	4.0		
	2.5		2440*1220		1.2		4.0	Specification	Thickness		2440*1220		1	Zinc Coated Steel SGCC			5.0
3.0					4.7	2440*1220	0.5	1.2	2440*1220	0.8		Brass					
3.2		AL1060			5.0		0.6	1.5		1.0		Specification	Thickness				
3.5		2440*1220	1.6	6.0	0.8		2.0	1.2		1500*3000		4.75	2.5				
4.0			3.0	0.1	1.0		SPHC				1.5						
5.0			3.5	0.15	1.2		Specification	Thickness			2.0						
6.0		3.5	0.2	1.5	1500*3000		6.0	3.0									
							0.4	2.3			9.7	4.0					
								3.0			Pickling Plate		Titanium Alloy TC4				
Brushed Plate					Brushed Plate		SUS430		2500*1250								
2440*1220	0.8	2440*1220	0.6	2440*1220	0.8		2440*1220	0.5	Specification	Thickness	Specification	Thickness	Specification	Thickness			
	1.0		1.0		1.0	4.0		4.5	1000*800	0.2							
	1.2		1.2		1.2	1.0			900*800	0.3							
	1.5		1.5		1.5	1.8			800*800	0.5							
	1.6		1.6		1.6	3.0											
2.0	2.0	2.0															
3.0	3.0	3.0															
AL6063		AL3003						SPTE									
Specification	Thickness	Specification	Thickness	Specification	Thickness	Specification	Thickness	Specification	Thickness	Specification	Thickness	Specification	Thickness				
2440*1220	1.5	2440*1220	2.0	2440*1220	0.8	2440*1220	1.0	2500*1250	4.0	2440*1220	0.8	1400*700	0.5				
	2.0		2.5		1.2		1.8		4.5		1.2						
		1220*1220	6.0														

3. Sheet Metal Processing Method List



4. Bending Radius Angle description

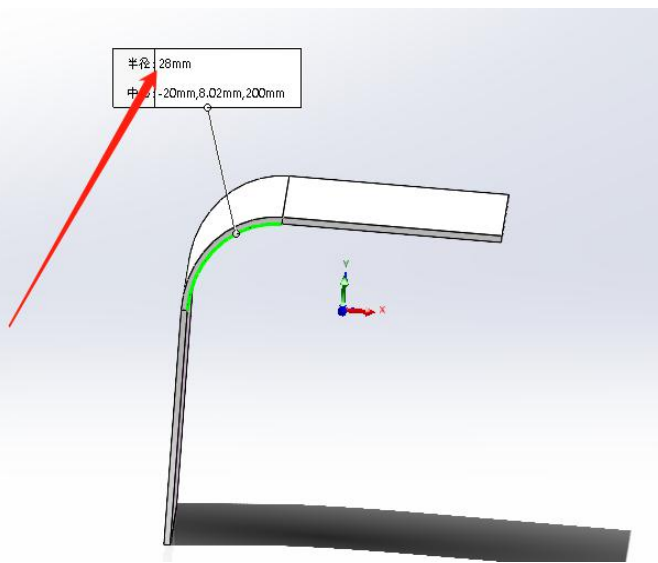


Bending Radius Angle Description

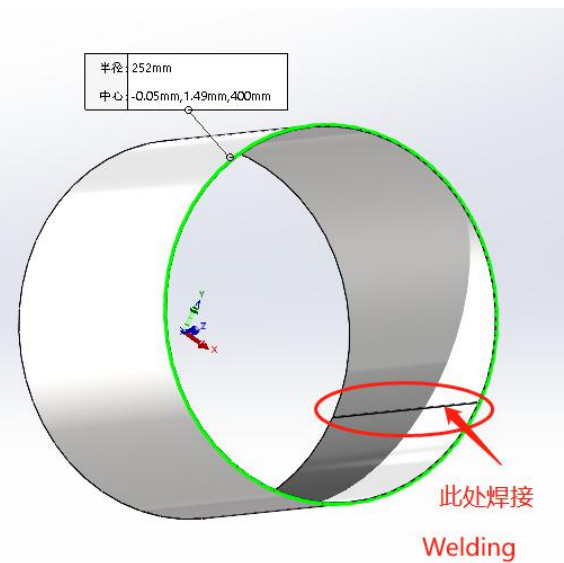
- There is no upper limit on bending angle, only lower limit. (Picture 1)
- The minimum bending radius is R0.5. (Inner radius angle).
- The thicker the bending material is, the greater the radius-angle will be. Vice versa.
- The bending radius of stainless steel is slightly larger than that of iron.
- The bending radius iron is slightly larger than that of aluminum.
- The R Angle below twice the plate thickness is prepared according to the standard of R0.5 in the normal bending tool, and angle R above twice the thickness of the plate is bent according to the arc.
- Sheet metal process can also produce large round tubes and conical products. (picture 2)



Picture 1



Picture 2



5. Processing Method



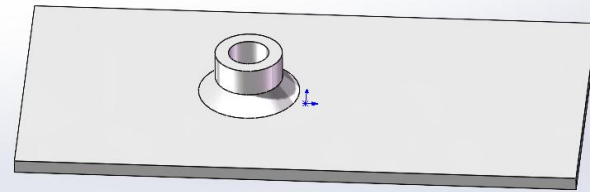
Bending Bridge



Bending gusset



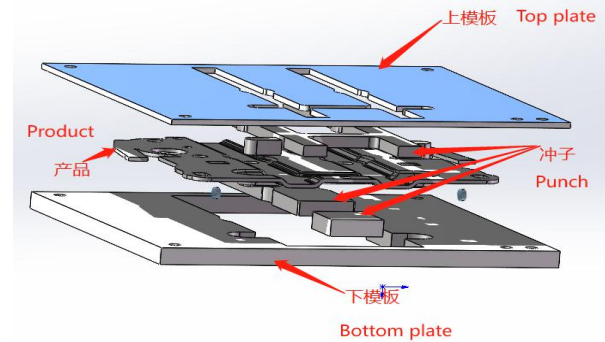
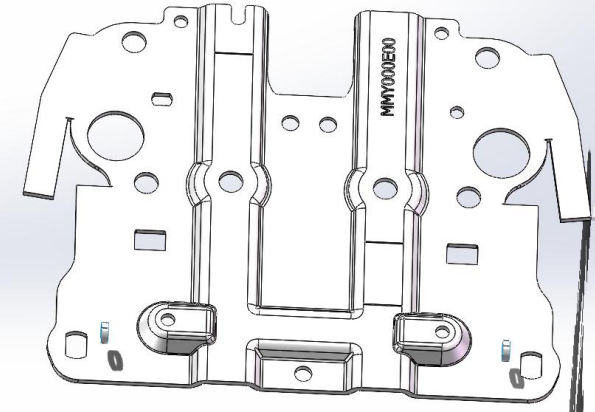
Embossing



Producing Protruding Point

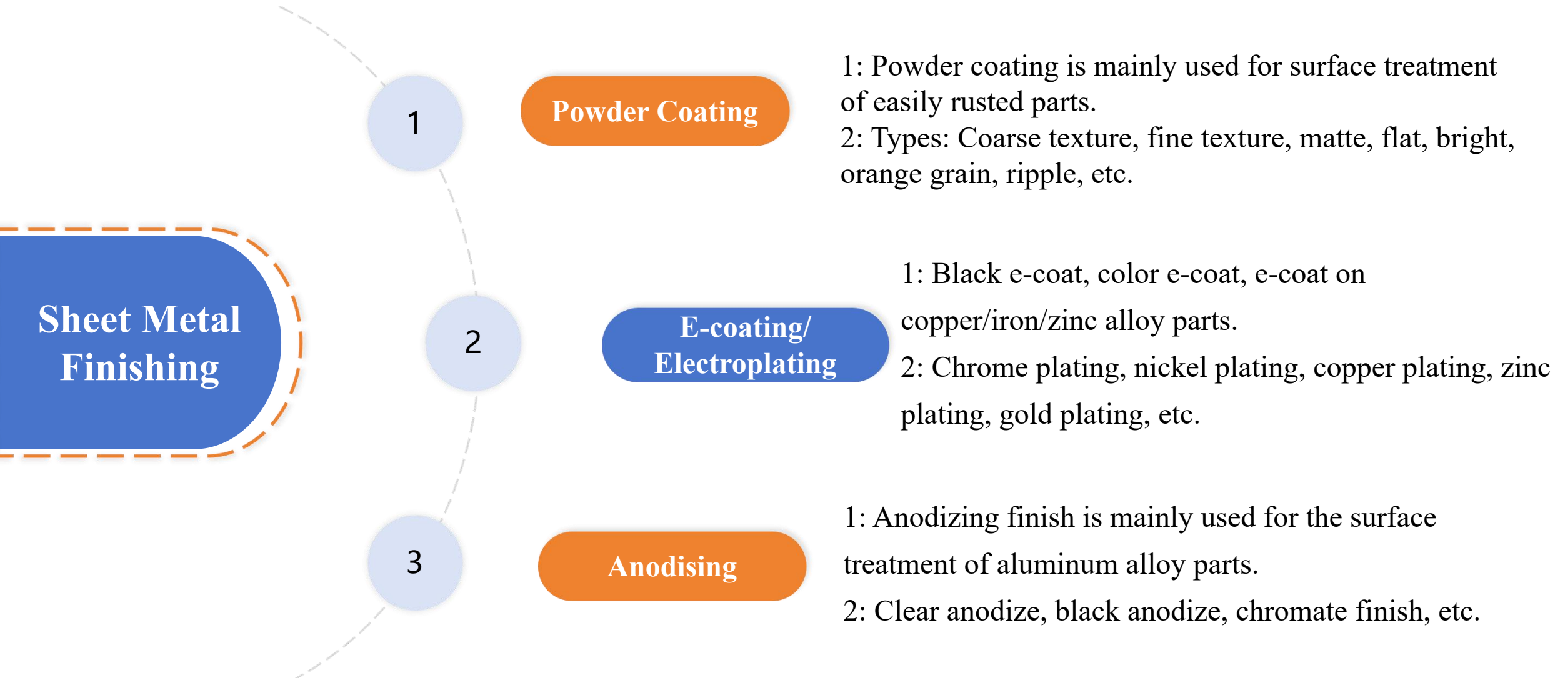


Processing Convex Hull



Producing simple tool

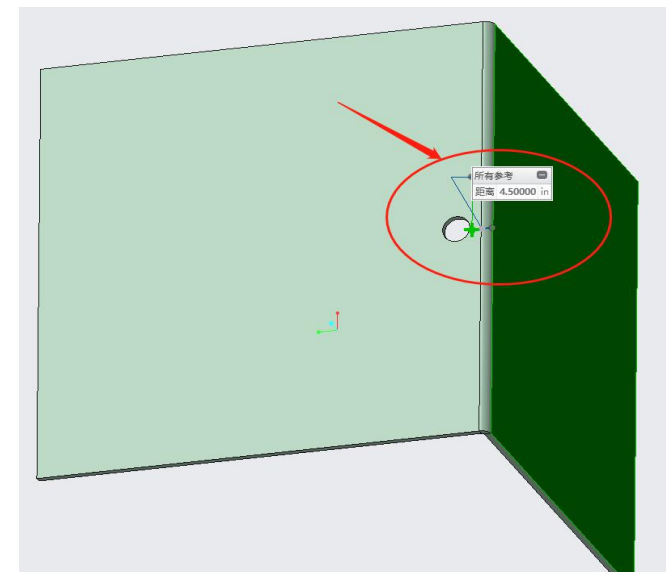
6. Sheet Metal Finishing



7. Distance from Bend Edge to Hole

Normal Bending Instructions

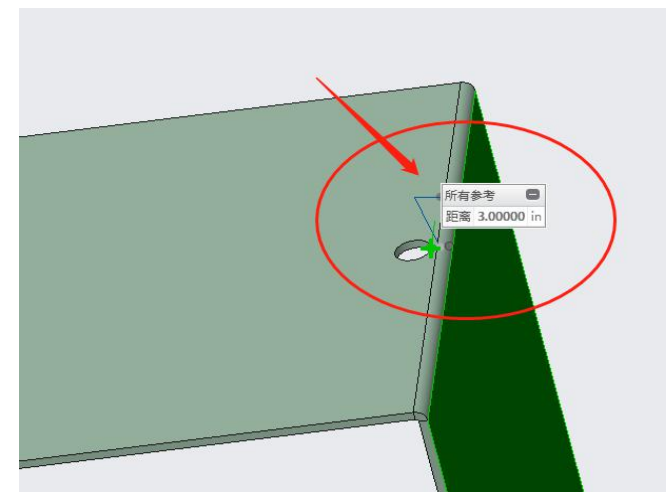
- The distance from the hole edge to the bend edge of 1.5mm plate in the normal bent state is 4.5mm. Refer to picture 1.
- A sheet metal part in the absence of any media or other auxiliary tools, in order to normal bend, the distance from the hole edge to the bending edge should be more than 3 times the thickness of the material.
- When the material is different, the corresponding hole edge to the bending edge of the distance will be different. Aluminum alloy 3 times the material thickness, iron 3.5 times the material thickness, stainless steel or relatively hard material need to be designed in 4 times the material thickness. Otherwise it can only use abnormal bending method, the cost will be relatively high.



Picture 1

Abnormal Bending Description

- The distance from the hole edge to the bent edge of 1.5mm plate in the abnormal bent state is 3mm. Refer to picture 2.
- In this case, it can be completed by the bending eccentric method, the cushion method, the line pressing method and the laser secondary repair method to complete.



Picture 2

8. Welding Process Specification

Hand-held Laser Welding Machine

- Suitable for welding thin parts.
- No need to polish, but the strength of the welding is not perfect.
- Not suitable for products with high welding requirements.



Argon Arc Welding Machine

- It can weld products within 15mm, with high strength, no welding scar and no flying slag.
- Slow efficiency and large product deformation.
- Suitable for thick copper, iron, aluminum stainless steel parts.
- It is commonly used and the frequency of use is relatively high.
- The requirements for tailor-welded products will be slightly higher.



Carbon-Dioxide-Arc-Welding-Machine

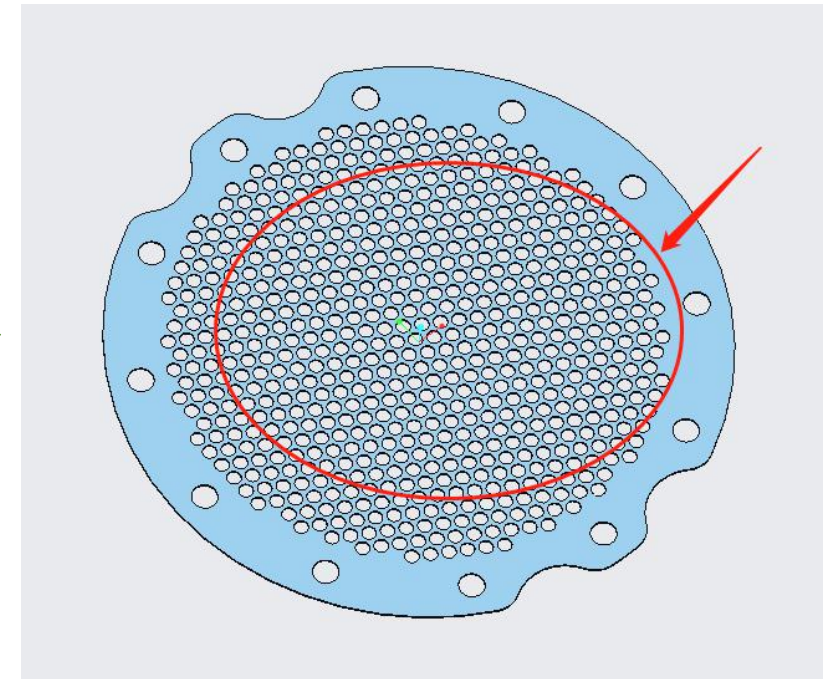
- Suitable for thick plates, quadrate pipe frame and products with large welds.
- High efficiency.
- With much flying slag, it's hard to clean.
- There will be deformation after welding.



9. Etching Process Analysis

Etching Process Analysis

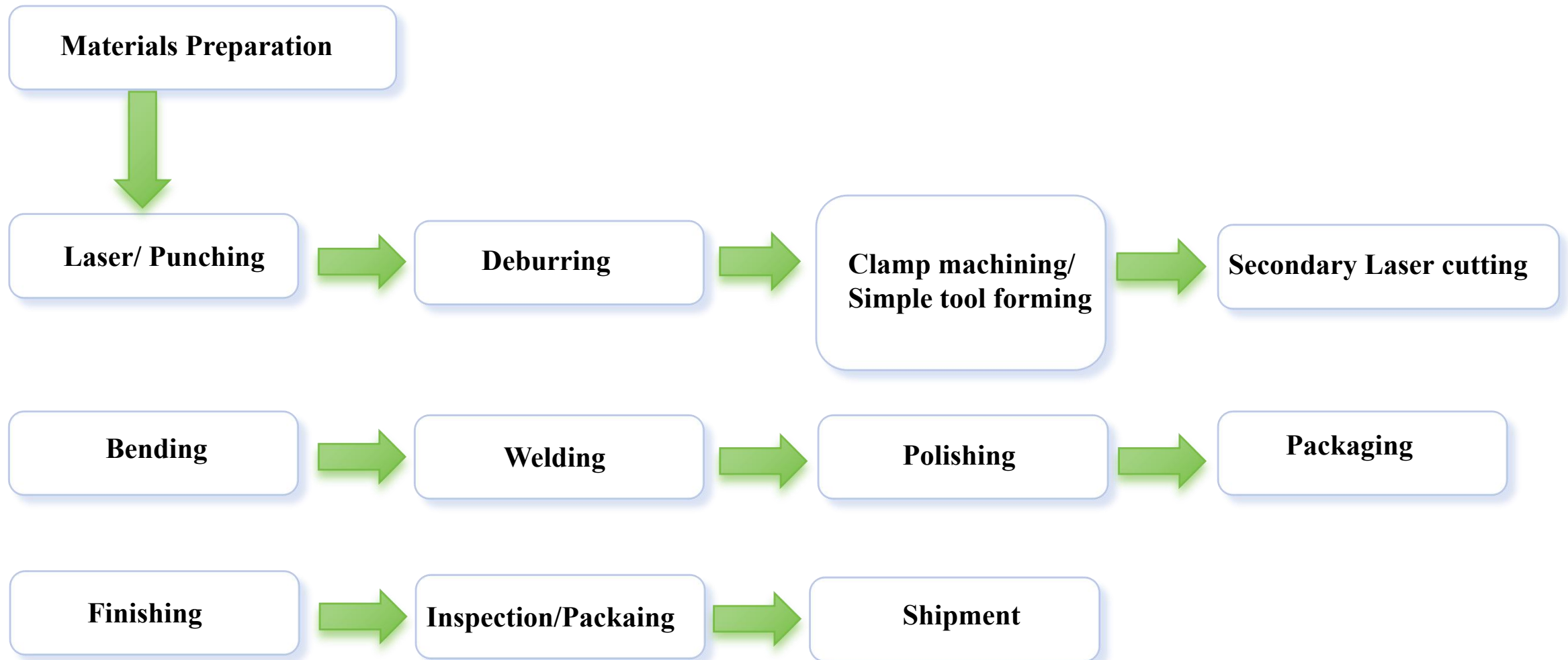
- As per the customer's requirements for the product to determine whether the sheet metal parts needs to be etched.
- Generally, the relatively thin and easy to deform plates are etched, such as plates below thick 0.12.
- Products with high flatness requirements also need to choose the etching process. Because laser cutting will lead to high temperature, it is easy to deform.
- As shown on the right, this workpiece is 0.2 thick material. Because it is cut mesh, all this type of shape will choose to use the etching process in order to prevent deformation.
- The most important thing is to determine the process according to the customer's product structure.



10. Propertie and Distinction of Raw Material

Types	Material	Properties	Finishing	NG
Aluminium alloy	1100/1060/3003/5052/6061/6063	1k series to 6k series, the larger the value is, the higher the hardness of the material have. Aluminum is lightweight, with good performance of heat absorption and fast heat dissipation	Sandblasting, anodizing, powder coating and plating	There may have pitting and the surface magbe easy to scratch.
Stainless steel	201/301/430/304/316	201 is the ordinary stainless iron magnetic, 301 is the shrapnel with weak magnetic, 430 is the mirror stainless iron with good magnetic, 304 and 316 are food grade stainless steel with non-magnetic.	Brushing, passivation, electrolytic grinding, powder coating	Easy to scratch
Iron	SPCC/SPHC/SECC/S GCC/SPTE	SPCC and SPHC belong to the bottom carbon steel cold plate with good ductility and easy rust, SECC belongs to the electrolytic plate with electrolytic layer on the surface, SGCC belongs to the galvanized plate with galvanized layer on the surface, and SPTE belongs to the tinplate with tin layer on the surface.	Powder coating, electrophoresis, electroplating	Easy to rust, need to do anti-rust treatment
Carbon steel	Q235/SPHC	This type of plate is a hot plate and is usually used for simple tool.	Powder coating	There will have pits in the surface
Copper alloy	Copper, brass, beryllium copper, phosphor bronze	Red copper belongs to pure copper, with good electrical. conductivity.Brass belongs to alloy copper.	Electroplating, oxidation, mirror polishing	Easily oxidized and discolored

11. Sheet Metal Processing Flow Chart



Thank you for watching

Zintilon



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